

Problem Setup

September 23, 2026

Problem Setup

We are given the quadratic equation

$$x^2 + 4x + 4 = 0$$

and we are asked to find all real values of x that satisfy it.

Strategy and Key Principles

The equation is a quadratic polynomial. We can solve it by:

- Factoring the quadratic, if possible, or
- Completing the square, which is equivalent to factoring for perfect-square trinomials.

Since the coefficients suggest a perfect square, we will complete the square.

Detailed Solution

$$x^2 + 4x + 4 = 0$$

$$x^2 + 4x = -4$$

$$\text{Add } (4/2)^2 = 4 \text{ to both sides} \implies x^2 + 4x + 4 = 0$$

$$(x + 2)^2 = 0$$

$$x + 2 = 0$$

$$x = -2$$

Final Answer

$$\boxed{x = -2}$$

Key Insights

- The quadratic $x^2 + 4x + 4$ is a perfect square: $(x + 2)^2$.
- When a squared term equals zero, the base of the square must be zero, giving a single (double) root.
- A common mistake is to forget to add the same constant to both sides when completing the square; here the constant is $(\frac{4}{2})^2 = 4$.